



Off-target Safety Analysis of Zinc Finger Proteins Targeting SNHG14 in Angelman Syndrome

Kai Barker
kaibarker@ucsb.edu

June 2026

1 Abstract

Angelman syndrome (AS) is a neurogenetic disorder arising from deficient UBE3A expression in the brain. Research has demonstrated the ability to suppress a gene, SNHG14, and restore paternal UBE3A expression in an AS mouse using zinc finger protein treatments. Zinc finger proteins targeting SNHG14, however, carry the risk of off-target gene disruption, motivating the need for safety classification. We applied PCA-based Pareto frontier analysis to identify candidate treatments, and integrated Open Targets disease associations with Nearest Shrunken Centroid feature selection to derive a subset of clinically relevant off-target genes. From this we computed off-target burden scores. We found that hATF561 exhibited the lowest overall off-target burden, while nZF139 exhibited the highest. This work provides a reproducible framework for off-target safety prioritization in zinc finger gene therapy candidate selection, functional for other target genes.

2 Introduction

Angelman syndrome (AS) is a rare neurogenetic disorder affecting approximately 1 in 15,000 births, characterized by severe developmental delays, impaired communication, and recurrent seizures. AS is caused by the loss of UBE3A expression in the brain, a gene that is active only from the maternal copy in neurons. A functional paternal copy exists but remains silenced by SNHG14, a long non-coding RNA. Suppressing SNHG14 releases this block and restores paternal UBE3A expression without introducing foreign DNA, making it a promising therapeutic target that avoids risks associated with direct gene editing approaches.

Zinc finger proteins (ZFPs) are engineered proteins that can be designed to suppress a specific gene, in this case SNHG14, without permanently altering the genome. O’Geen et al. (2023) designed 63 zinc finger sequences, of which 11 were selected for testing based on proximity to the target site. A single injection of the lead construct partially restored motor deficits in AS mice. However, no treatment is perfectly precise, and treatments can also disturb other genes, which may carry associations with unintended health consequences across a range of diseases. Selecting the best candidate therefore requires not only measuring how well each treatment suppresses SNHG14, but also characterizing its off-target effects.

We aimed to characterize which of the treatments were optimal, in terms of suppression of the SNHG14 gene and the number of off-target genes disturbed, as well as investigating the disease-linked disruptions to off-target genes when applying these treatments. We applied PCA and a Pareto frontier analysis to identify top candidate treatments, and then used a Nearest Shrunken Centroids (NSC) algorithm to identify the genes most relevant to distinguishing treat-

ments from control groups. From this, we constructed off-target burden scores by combining disease association evidence from the Open Targets dataset with the amount that the disturbed genes were changed.

Our treatments dataset was provided by the Segal Lab at UC Davis, whose published research on zinc finger constructs for AS treatment motivated this analysis. Discussions with Dr. Segal identified off-target safety characterization as an open question to his lab. This project applies large-scale computational methods to address it by combining gene expression data with the external Open Targets disease association database.

3 Data

The dataset used in this analysis combines two separate sources. The first was gene expression data collected and provided by the Segal Lab at UC Davis (Segal, 2026), measuring the expression of 78,986 genes across 48 samples representing 22 zinc finger treatments (11 sequences with 2 replicates each) and a control (2 replicates), applied in human cells. We also used data sourced from Open Targets (Carvalho-Silva et al., 2019), a public database aggregating genetic and biomedical evidence for gene-disease associations, which was accessed via their public Google BigQuery platform.

To construct our working dataset, we joined the Open Targets `association_overall_direct` and `disease` tables on disease ID, filtering to gene-disease combinations with an association score greater than 0.1 to retain only meaningfully supported associations. This left a parquet file with 705,265 rows and 6 columns: `gene_id`, `disease_name`, `disease_id`, `therapeutic_area_ids`, `association_score`, and `therapeutic_areas`. Each row represents a unique gene-disease association pair, covering approximately 18,000 of the original 78,986 genes that had qualifying Open Targets evidence. Each disease record also includes therapeutic area classifications (broad categories such as cardiovascular disease or nervous system disease).

It is important to note that “disease” in the Open Targets platform describes four categories: diseases, phenotypes, measurements, and biological processes. Association scores quantify the strength of evidence linking a gene to a disease, aggregated across multiple data sources including genetics, literature, and other databases. Each data source is scored independently using a harmonic sum of its underlying evidence scores, which are combined into an overall direct association score between 0 and 1, where higher values reflect stronger evidence across sources (Open Targets, 2024).

4 Methodology

Our analysis aimed to identify which zinc finger treatments optimally balanced SNHG14 suppression against off-target gene disruption, and to characterize the disease-linked implications of those disruptions.

We first applied PCA to the full gene expression matrix, finding that 36 principal components captured approximately 80% of the variance. For each treatment and control, we averaged across two replicates and computed the Euclidean distance to the control cluster center in 36-dimensional space. Treatments closer to the control are less disruptive. We then constructed a Pareto frontier using $\log_2$ fold change of SNHG14 as the x-axis (power or efficacy) and Euclidean distance to control as the y-axis (safety). Treatments on the frontier represent the optimal tradeoff, where no other treatment achieves greater target gene suppression without also increasing off-target disruption. Four treatments were identified as Pareto-optimal.

To identify the specific genes driving the differences between these optimal treatments and control samples, we framed the analysis as a binary classification problem. Given the high-dimensional nature of the dataset relative to the relatively small number of samples (4 Pareto-optimal treatments and 1 control, with 2 replicates each), traditional techniques like LASSO were insufficient. Since LASSO is constrained to select at most n (number of rows) features, it was unsuitable for such a problem. Instead, we applied Nearest Shrunken Centroids (Tibshirani, Hastie, Narasimhan, & Chu, 2002), a method specifically designed for high-dimensional genomic classification. NSC computes a standard centroid for each class, and applies a soft shrinkage penalty to pull class centroids toward an overall mean. Genes whose centroids are shrunk entirely to the global mean are discarded, effectively performing feature selection and isolating the genes that most meaningfully distinguish the Pareto-optimal treatments from the controls.

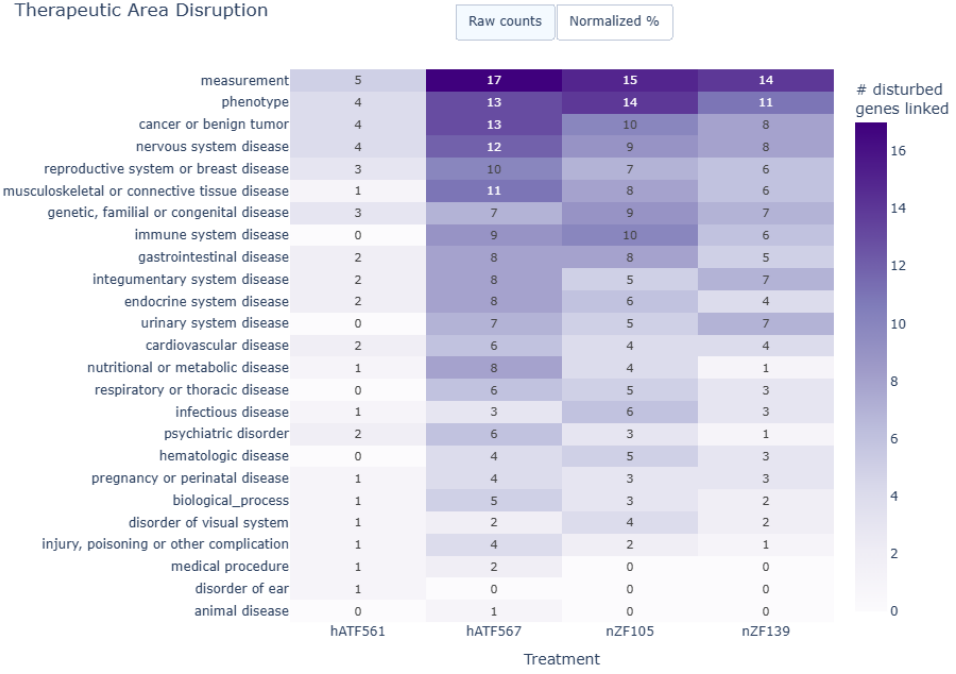
To assess the genes selected by NSC, we performed bootstrap resampling with 50 iterations, sampling within treatment and control groups separately, given the small number of control replicates. This was applied across shrinkage thresholds of 0.5 to 3.0, with fine-tuning between 2.0 and 2.5 where significance was highest, and used permutation testing (100 permutations per threshold) to validate that selection rates were beyond random chance. NSC selected genes were placed into 3 tiers for those selected at 100% frequency across all thresholds, those selected at threshold 2.4 (most significant yet tightest threshold), and those at the broadest threshold of 0.5.

For each Pareto-optimal treatment, we then computed off-target burden scores across all gene tiers. Burden was calculated as the sum of $|\log_2 FC| \times \text{association score}$ across all genes exceeding a fold change threshold of 1.0 (also filtered for those in the Open Targets dataset). We applied a $|\log_2 FC|$ threshold of 1.0 as it

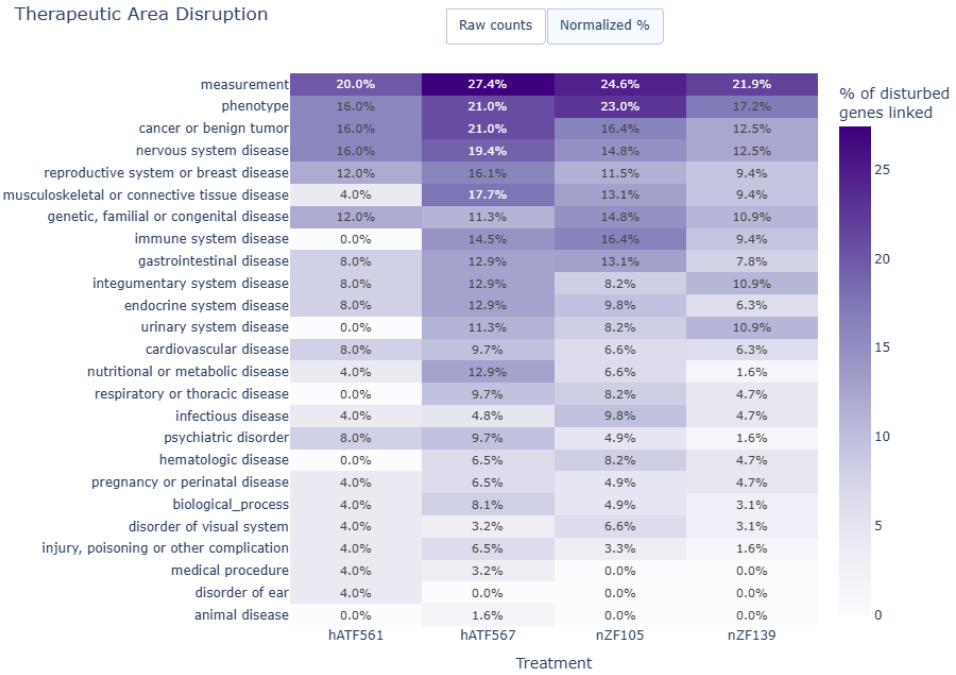
corresponds to a minimum two-fold change in gene expression. While Love, Huber, and Anders (2014) caution against hard fold-change filters as a substitute for statistical inference, they do note that changes of sufficient magnitude are necessary to distinguish biologically significant from random chance variation. They also use the same threshold as a benchmark in their analysis. In our analysis, this threshold serves as a noise filter to concentrate our burden scores on substantial disruptions, rather than as a significance test. Burden was further split into shared components (genes selected by NSC across treatments) and treatment-specific components, providing a measure of whether the off-target disruption reflects a class-wide effect of the treatments or individual behavior of specific treatments.

5 Results

We first examined which genes showed substantial expression changes across the four Pareto-optimal treatments. Filtering to genes with $|(\log_2 FC)| \geq 1.0$ and Open Targets disease evidence yielded the following counts: hATF567 (62 genes), hATF561 (25 genes), nZF105 (61 genes), and nZF139 (64 genes), summarized in Figure 1. hATF561 showed the fewest disease-linked disturbed genes across all treatments, while nZF139 showed the most.



(a) Raw counts



6
(b) Normalized

Figure 1: Therapeutic area heatmaps of disease-linked disturbed genes ($|\log_2 FC| \geq 1.0$) across Pareto-optimal treatments.

These genes were then mapped against the Open Targets disease associations to examine which therapeutic areas were most affected. Figure 1 shows the resulting heatmap of gene counts by therapeutic area across treatments. The measurement and phenotype categories account for the largest share of associations, and notably, the measurement category alone contains 4,651 unique entries, phenotype 8,178, and biological process 202. While their dominance is somewhat reassuring, as the size of these categories reflects a broad range of associations rather than specific serious diseases, the more clinically concerning categories such as nervous system disease and cancer or benign tumor demand closer attention in the safety evaluation of these treatments. hATF561 shows the smallest overall footprint across all categories, while nZF139 and hATF567 show the broadest.

We grouped our NSC identified genes into three tiers based on bootstrap stability: 7 most-robust genes selected in 100% of bootstraps across all thresholds up to 2.5, 22 genes at the most strict threshold that maintained high significance (threshold 2.4, $p \leq 0.001$), and 1,484 genes at the broadest threshold of 0.5. At threshold 3.0, no genes were selected in 100% of bootstraps. The threshold range between 2.0 to 2.5 was chosen for additional refinement as the transition from highly confident ($p \leq 0.001$) to more moderate significance ($p = 0.030$ at 2.5) occurred. These three tiers serve different analytical purposes, and one should not be thought of as superior to another. The most robust genes offer maximum confidence for an in-depth follow up, the 22 balance coverage with control, and the broad 1,484 genes capture the largest most stable signal. The appropriate set should depend on intended analysis and application. Since cross-validation could not be used for threshold selection due to the small sample size, the chosen thresholds are more not proven to be statistically optimal, and should be treated approximate rather than definitive.

NSC-Selected Gene Expression (threshold=2.4, 22 genes)

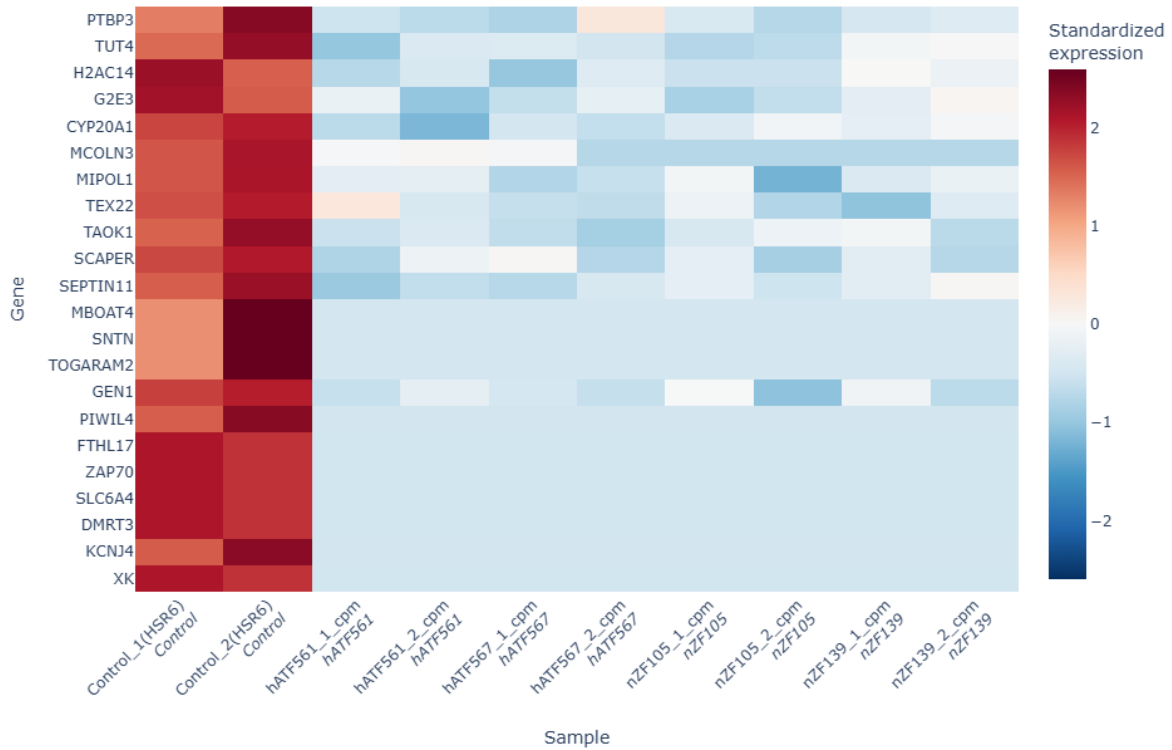


Figure 2: NSC-Selected Genes at the 2.4 threshold, gene names against treatment and control replicates. Colors reflect z-scores computed per gene across all samples. Red indicates expression above the gene's mean, blue indicates below.

Genes appearing consistently red across control samples in and blue across treatment samples in Figure 2 indicate that NSC-selected genes are more highly expressed in controls and consistently suppressed with treatments. This pattern suggests that in addition to suppressing SNHG14, the ZFPs broadly lower the expression in a set of genes with meaningful disease associations.

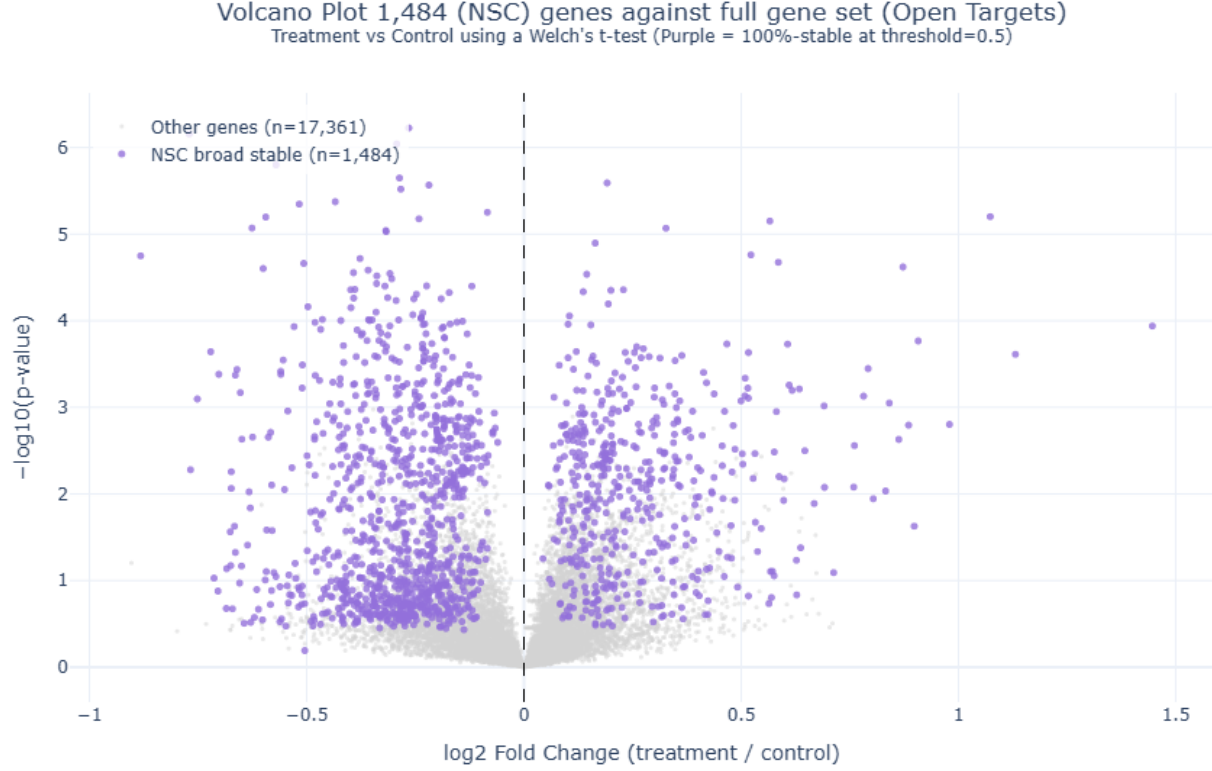


Figure 3: Volcano Plot of the 1,484 NSC broad threshold genes (100% bootstrap stability at threshold 0.5). Purple points indicate broad signature genes, gray points represent the remaining Open Targets genes. Y-axis is inverted with a negative log to show increasing p-value from Welch's t-test.

Figure 3 shows that the NSC-selected genes tend to cluster at higher significance levels on the volcano plot, indicating their expression is more strongly differentiated between treatment and control conditions. Welch's t-test was used rather than Student's t-test as it does not assume equal variances between groups, making it more appropriate given the small and unequal sample sizes between treatments and controls. We see that the purple points have the largest spread across the most extreme $\log_2$ fold changes, consistent with their selection as the most discriminative genes.

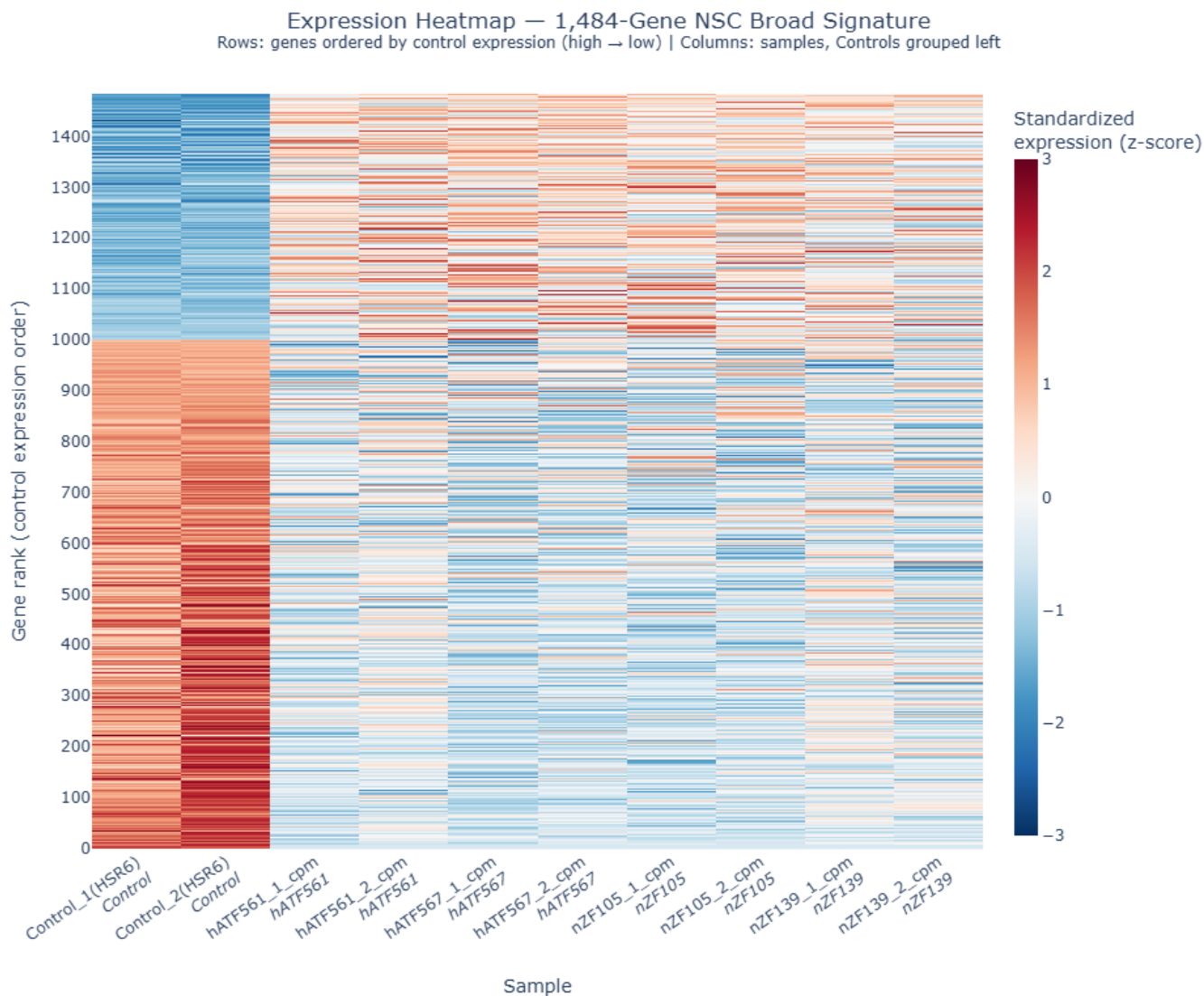


Figure 4: Heatmap of the 1,484 NSC broad signature genes. Rows represent genes ordered by standardized expression in control samples (highest to lowest), columns represent individual samples with controls grouped on the left. Colored by z-scores computed per gene across all samples.

The broad signature heatmap shows that of the 1,484 NSC-selected genes, approximately 400 show below-average standardized expression in control samples (blue) while the remaining $\sim 1,000$ show above average expression (red). Across

all four treatments and their replicates, a general gene expression inversion is apparent. Genes highly expressed in controls tend to be suppressed under the treatments, and vice versa. This pattern is again consistent with the NSC selection criterion, as these genes were chosen because their expression distinguished treatment from control. While the heatmap confirms this trend visually, it is important to note that the inversion is not uniform across all genes, with 1,484 rows, the visualization is best interpreted as a general pattern rather than a precise analysis.

Using the NSC-selected genes, we were able to compare these again with our Open Targets enriched data.

Table 1: Top diseases associated with ≥ 2 NSC-selected genes (threshold = 2.4, 22 genes)

Disease	N Genes	Avg Score	Max Score
Neurodegenerative disease	10	0.365	0.557
Body height	8	0.317	0.512
Platelet volume	7	0.358	0.521
Erythrocyte count	6	0.370	0.507
Body mass index	6	0.362	0.515
Sitting height measurement	6	0.349	0.499
Eosinophil count	6	0.339	0.539
Serum creatinine amount	6	0.298	0.476
Red cell distribution width	6	0.285	0.355
High density lipoprotein cholesterol measurement	6	0.233	0.404
Blood protein amount	6	0.208	0.310
Erythrocyte volume	5	0.389	0.517
Body weight	5	0.301	0.467
Total cholesterol measurement	5	0.294	0.471
Platelet count	5	0.259	0.420
IGF-1 measurement	5	0.246	0.391
Genetic disorder	4	0.386	0.527
Lean body mass	4	0.349	0.503
Hemoglobin measurement	4	0.295	0.461
Mean reticulocyte volume	4	0.284	0.458

Table 2: Therapeutic area associations of NSC-selected genes (threshold = 2.4, 22 genes)

Therapeutic Area	N Genes	N Diseases	Avg Score
Measurement	19	207	0.280
Nervous system disease	18	70	0.354
Phenotype	13	69	0.331
Genetic, familial or congenital disease	12	40	0.348
Reproductive system or breast disease	11	17	0.239
Cancer or benign tumor	11	21	0.232
Musculoskeletal or connective tissue disease	10	21	0.260
Psychiatric disorder	8	37	0.372
Gastrointestinal disease	8	17	0.261
Endocrine system disease	7	16	0.280
Integumentary system disease	6	5	0.272
Immune system disease	6	19	0.357
Disorder of visual system	5	9	0.253
Infectious disease	5	11	0.309
Nutritional or metabolic disease	4	7	0.359
Biological process	4	4	0.326
Urinary system disease	4	7	0.268
Injury, poisoning or other complication	3	4	0.217
Pregnancy or perinatal disease	3	2	0.245
Cardiovascular disease	3	9	0.282
Hematologic disease	2	2	0.187
Respiratory or thoracic disease	2	6	0.236
Medical procedure	1	1	0.233
Disorder of ear	1	1	0.185

Using the 22 NSC-selected genes (threshold 2.4), we characterized their disease associations using the Open Targets data, shown in Tables 1 and 2. Neurodegenerative diseases have the most associated selected genes, with an average association score of 0.36, and a maximum of 0.56. Notably, the second most associated disease by gene count is body height, which represents an interesting direction for future work looking into the implications of disrupting these genes. Consistent with the earlier therapeutic area analysis from Figure 1, phenotype and measurement remain among the top three therapeutic areas. Nervous system disease has 18 associated genes across 70 diseases in the category, showing the relevance of understanding the off-target effects of the selected genes. The frequency of neurodegenerative disease associations at this tier of selection is notable given that Angelman syndrome itself is a neurological disorder. While the significance and direction, whether the associations are positive or negative, cannot be determined from our data alone, the concentration of neurologically relevant genes in this set suggests needed caution and further research into these treatments.

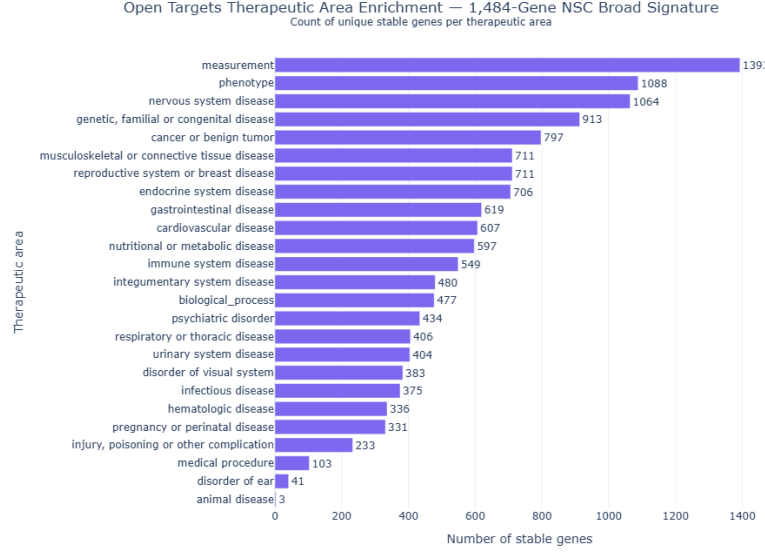


Figure 5: Distribution of the 1,484 NSC genes (threshold = 0.5, 100% bootstrap stability) across Open Targets therapeutic areas.

Figure 5 shows the distribution of the 1,484 genes across Open Targets therapeutic areas. Consistent with our analysis of the threshold 2.4 genes, measurement and phenotype represent the two largest categories. More notably, over 1,000 broad signature genes are associated with nervous system diseases and between 700 and 900 are associated with reproductive, cancers or tumors, and congenital diseases. Given that these genes are systematically disturbed across all Pareto-optimal treatments, their association with neurological and cancer-related disease pathways warrants careful consideration in the safety evaluation of any ZFP targeting SNHG14.

These findings indicate that Pareto-optimal ZFP treatments are not equal in terms of off-target effects. While ranking the risk of individual diseases is inherently subjective, we constructed an off-target burden score based on the number of disease-linked disturbed genes weighted by their Open Targets association scores, providing a numerical basis for comparison.

Table 3: Per-treatment off-target risk metrics for Pareto-optimal treatments, ranked by composite risk score (highest to lowest). Disease-linked genes are filtered for $|\log_2 FC| \geq 1.0$

Treatment	Disturbed genes	Disease-linked genes	Burden (weighted)	High-conf. disrupt.	Ther. Areas touched	Composite rank
hATF567	62	17	236.98	1	24	3.500
nZF139	64	14	138.99	2	22	3.375
nZF105	61	18	130.41	0	22	2.000
hATF561	25	6	43.57	0	20	1.125

Table 3 explains the per-treatment metrics. hATF567 carries the largest off-target burden at 236.9 across 62 disturbed genes, while hATF561 has the smallest burden of 43.57 with only 25 disturbed genes.

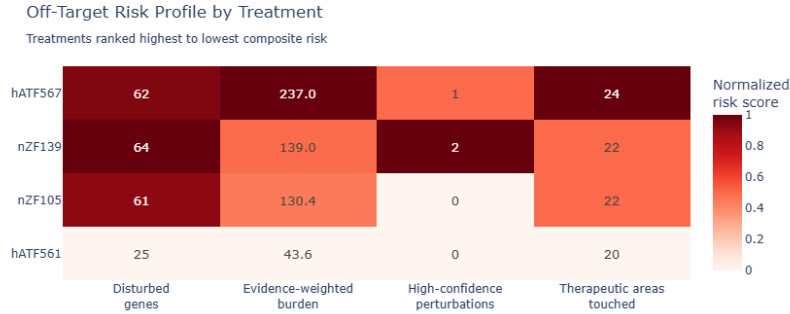


Figure 6: Off-Target Risk Profile by Treatment. Each section is given a normalized risk score from 0 to 1, so coloring is affected only within that category, not throughout the entire heat map.

Figure 6 normalizes these metrics within each category for visual comparison, showing that hATF561 scores the lowest across all four risk areas.

Table 4: Evidence-weighted burden decomposed into shared (NSC-selected) and treatment-specific components.

Treatment	Shared genes	Shared burden	Specific genes	Specific burden	Total burden	% Shared
hATF567	12	134.16	5	102.82	236.98	56.61
nZF139	8	24.25	6	114.74	138.99	17.45
nZF105	13	82.06	5	48.35	130.41	62.93
hATF561	1	1.01	5	42.56	43.57	2.31

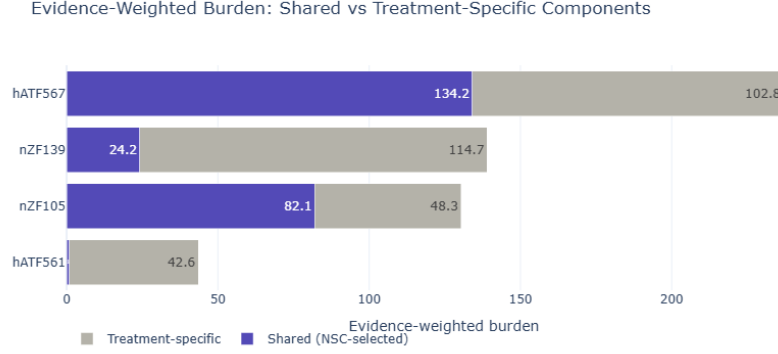


Figure 7: Evidence-weighted burden decomposed into shared (NSC-selected, purple) and treatment-specific (gray) components for each Pareto-optimal treatment.

Table 4 and Figure 7 split the burden into shared and treatment-specific components. hATF561 continues to show the smallest overall burden, with only 1 point contributed by shared genes, meaning its off-target footprint does not reflect the broader class-wide disruption pattern seen across treatments, further strengthening its position as the safest candidate by this metric. nZF139 shows the next lowest shared burden but the second highest overall burden, driven almost entirely by treatment-specific disruption, suggesting its off-target effects are unique to this treatment rather than a general consequence of the ZFPs. hATF567 carries the highest burden in both components, contributing to its highest overall score.

Taken together, these results consistently identify hATF561 as the Pareto-optimal treatment with the lowest off-target burden across all metrics and tiers, while nZF139 and hATF567 present the most concerning safety profiles by this approach.

6 Conclusion and Future Study

Our work intended to characterize which zinc finger protein treatments were optimal, both in terms of suppression of the target SNHG14 gene and in reducing the number of off-target genes disturbed. Suppressing SNHG14 restores expression of the paternal UBE3A allele, which is silenced in Angelman syndrome. We applied a PCA and Pareto-frontier analysis to identify which ZFP treatments were optimal, and were left with four candidates: hATF561, hATF567, nZF105, and nZF139.

These four Pareto-optimal treatments along with their replicates and the two

control replicates were used as the response in a Nearest Shrunken Centroids supervised learning algorithm to identify genes whose change in expression over the sample space contributed to the classification. We identified three main tiers of meaningfully disturbed genes, with each serving a different purpose based on amount and robustness. We found 1,484 genes at the least robust broad threshold, 22 at the most refined yet highly significant threshold, and 7 at the most strict threshold. These thresholds were all tested for significance with a permutation test. Apache Spark was used to fit these NSC models across all permutations for every threshold, minimizing compute time over 1,800 models.

We found that across the four Pareto-optimal treatments, genes selected by NSC tended to be more significant compared to those not selected (Welch’s t-test), and had a larger spread across $|\log_2 FC|$ (fold change). Additionally, when the 2nd tier gene expressions were standardized for all observations, control observations had a higher than average standardized expression, which was suppressed across all genes with ZFP treatments. In the most broadly selected genes, a majority ($\sim 70\%$) of control observations had an above average expression, with the rest having below average gene expression. A general trend of reversal was noticed with the ZFPs either suppressing or increasing gene expression when applied. These four Pareto-optimal treatments were assigned general risk scores based on their burden, a calculation based on the disease associated genes they disturbed and the value of that association score. hATF561 was identified as having the lowest overall burden score and hATF567 the highest.

Our findings suggest that the ZFP treatments were not all equal, as only 36 of nearly 80,000 PCs explained roughly 80% of variance. Of these Pareto-optimal treatments, they tend to each disturb a range of off-target genes associated with a number of therapeutic areas such as cancers, neurological disorders, and physical attributes (measurements). This suggests that while the treatments could suppress SNHG14, a promising route for restoring UBE3A in AS, there are off-target risks associated with these treatments, even in our identified least-burdensome treatment. Future work would benefit from aiming to classify one disease as more dangerous or harmful compared to another. If a database were accessible to indicate that body weight association was a more acceptable association compared to breast cancer, a better system for off-target risks could be created. Further, analyzing all available treatment burdens would show whether the Pareto-frontier truly selected optimal treatments in terms of risk, and identifying whether all low-burden treatments meaningfully suppressed SNHG14 to a level needed to restore UBE3A expression. The NSC algorithm could also be improved with additional control samples, as our current sample size was incompatible for using any type of cross-validation to select optimal thresholds, and thus our current thresholds are less than statistically optimal. More permutations or bootstraps could be used as well, or the NSC approach could be compared to a DESeq2 (a different gene expression analysis method) approach which would provide adjusted p-values per gene and more reliable $|\log_2 FC|$ values for low-expressed genes.

As zinc finger protein therapies continue to advance towards clinical application, stronger risk classification methods will become increasingly important. This work demonstrates that integrating large-scale gene expression data and known disease association databases provides a reproducible and interpretable approach for further treatment prioritization, ideally extensible to other targets beyond SNHG14.

7 References

- Carvalho-Silva, D., Pierleoni, A., Pignatelli, M., Ong, C., Fumis, L., Karamanis, N., ... Dunham, I. (2019). Open targets platform: new developments and updates two years on. *Nucleic Acids Research*, 47(D1), D1056–D1065. doi: 10.1093/nar/gky1133
- Love, M. I., Huber, W., & Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology*, 15(12), 550. doi: 10.1186/s13059-014-0550-8
- O’Geen, H., Beitnere, U., Garcia, M. S., Adhikari, A., Cameron, D. L., Fenton, T. A., ... Segal, D. J. (2023). Transcriptional reprogramming restores UBE3A brain-wide and rescues behavioral phenotypes in an Angelman syndrome mouse model. *Molecular Therapy*, 31(6). doi: 10.1016/j.ymthe.2023.01.013
- Open Targets. (2024). *Open targets platform: Association scores*. Retrieved from <https://platform-docs.opentargets.org/associations> (Accessed 2026)
- Segal, D. J. (2026). *Zinc finger construct gene expression dataset for Angelman syndrome therapeutic screening*. (Data provided by the Segal Lab, University of California, Davis)
- Tibshirani, R., Hastie, T., Narasimhan, B., & Chu, G. (2002). Diagnosis of multiple cancer types by shrunken centroids of gene expression. *Proceedings of the National Academy of Sciences*, 99(10), 6567–6572. doi: 10.1073/pnas.082099299

8 Appendix

All code for this analysis is available at the following GitHub repository:

<https://github.com/kbarker223/Genetic-Expression-Analysis>

The author's primary contributions are `opentargets_analysis.ipynb`, `NSC_risk_analysis.ipynb`, and ETL functions in `load_data.py`